Incremental Predictive Value of Two Antenatal Visits for Recorded Low Birth Weight: Technical Report

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Generated by `python -m antenatal technical-report` on 2026-10-24. Every number is read from `outputs/`, `docs/evidence/` or `StudySpec`; none is typed by hand. Development results are internal validation; the held out partition was evaluated once, on 2026-10-24, on models locked beforehand.

1. Scope alignment

The proposal fixed the question, the cohort, the partition, the feature sets, the evaluation and a prototype before any model was fitted. The table maps each planned component to its status and evidence.

Planned feature	Purpose	Status	Evidence
Data loader with checksum; cleaning and eligibility (R1)	One exact file; Table 3.5 with a reason per excluded record	Completed	`outputs/tables/cohort_flow.csv`
Partition writer (R3) and guard	Seal the held out partition before any model step; refuse it in every fit	Completed; integrity audit 11 of 11	`docs/evidence/holdout_integrity.md`
Feature sets A, B, C (R2) and nested cross validation	The primary design on development records	Completed	`outputs/tables/feature_set_comparisons.csv`
Sensitivity analyses S1 to S6 and extensions	Test the comparison under changed rules	Completed	`outputs/tables/sensitivity_analyses.csv`
Prespecified improvement experiments (Stage 2)	Try to improve the model under a rule fixed in advance	Completed: 28 runs, 0 passed	`configs/experiment_plan.yaml`, `outputs/tables/experiments_v2.csv`
Statistical analysis of the research questions	Estimates, intervals, p values, Holm adjustment	Completed	`notebooks/03_statistical_analysis.ipynb`, `docs/research_results.md`
Held out evaluation (R5)	Evaluate the locked model once, logged	Completed once	`docs/holdout_results.md`, `partitions/holdout_access.log`
Model serialiser (R12, R16) and label (decision D6)	One artefact with checksum; a label with its confidence	Completed	`webapp/model/version.json`, `webapp/model/classification.json`

Planned feature	Purpose	Status	Evidence
Data model (Section 3.4.4)	Records, visits, configurations and results	Completed in SQLite; the budget named PostgreSQL	`python -m antenatal load-db`
Interface screens 1 to 5 (Table 3.10), entry rules (R9, R10), sign in (decision D1)	Register, two visits, assessment, caseload, with roles	Completed in Next.js over the API	`frontend/src/`, `docs/evidence/input_tests.md`
Cross browser, accessibility, performance, container, online host	Evidence and deployment	Completed; hosted online with a local backup (decision D7)	`docs/evidence/`, `Dockerfile`, `fly.toml`

2. Technical functionality and algorithms

Data and cohort. The released file (8,817 records) is read only after its SHA 256 matches the frozen value. Pure parsers turn text cells into typed values: blood pressure through a regular expression, sentinel tokens such as "not_checked" into missing values, and unambiguous numeric slips repaired. Values outside the physiological ranges of `StudySpec` become missing. `CohortBuilder` applies one named method per eligibility step and records the reason for every excluded record, giving the eligible cohort of 8,161 and the analysis cohort of 3,908 pregnancies with 206 low birth weight births.

Grouped stratified partitioning. Records that share a registration profile (age, last menstrual date, height, registration weight and prior deliveries) form a group, and whole groups are assigned to folds while balancing events. The first fold is the held out partition of 782 pregnancies with 41 events, written with a checksum before any model step. Every model fit carries a `TrainingGuard` that refuses any held out record, and the held out outcomes are read only through one logged gate. The development partition holds 3,126 pregnancies with 165 events.

Feature sets. Set A holds registration characteristics, set B adds the visit two measurements, and set C adds the visit one value of each measurement recorded at both visits for at least 60 percent of records, a screen refitted in every training fold. Linear models receive values only; tree models also receive the change between visits. Sets are subclasses of one abstract `FeatureSet`.

Models and validation. 8 model families and a constant prevalence baseline. Pipelines impute medians with missingness indicators and scale where needed; gradient boosted trees handle missing values natively. 5 outer folds repeated 10 times, with hyperparameters chosen in 3 grouped inner folds by average precision, the area under the precision recall curve. Predictions are averaged per pregnancy across repeats, and paired differences between feature sets on the same pregnancies get a group bootstrap interval from 2,000 resamples. The smallest detectable difference is 2.8 bootstrap standard errors (five percent two sided test, 80 percent power).

Prototype. A FastAPI application with routers that validate, services that decide and a database of synthetic cases. Scoring goes only through the frozen artefact, whose checksum is verified at start; a changed file stops the server. The interface shows rank, probability with an interval, and the label "Likely" or "Not likely low birth weight" at the threshold fixed at the highest 20 percent of out of fold predictions, with how often that label is right.

3. Testing approaches and results

The test suite ran 315 tests in 330.3 seconds with 0 failures: 242 unit tests, 67 integration tests and 6 browser tests. Line coverage of the package and the application was 97.0 percent of 3,789 statements. Ruff reported no findings and black reported no changes needed.

Approach	What it checks
Unit tests	Parsers on odd values from the file; each cohort count; partition disjointness, sealing and tamper detection; each metric against scikit-learn or a hand computation
Leakage and integrity	No held out record in any saved prediction; the guard refuses held out records; the served model rebuilds from development records to 1e-9
Integration tests of the API	Every refusal path, sign in and roles, a tampered artefact refusing to start, equality of API, service and package to 1e-9
End to end and browser tests	The full command sequence on a throwaway project; the five screens in three browsers at two sizes
Input variation	12 inputs entered through a running server

The input variation run gave 12 of 12 inputs behaving as expected:

No.	Input	Expected	Observed status	Result
1	Typical complete case	201, assessment returned	201	pass
2	Synthetic flag not set	422, message contains "synthetic flag"	422	pass
3	Age 121	422, message contains "Maternal age must be between 12 and 55"	422	pass
4	Age 55, upper boundary	201, assessment returned	201	pass
5	Height missing	422, message contains "Height is required"	422	pass
6	Visit one systolic 300	422, message contains "Systolic blood pressure must be between 60 and 250"	422	pass
7	Visit one diastolic 20	422, message contains "Diastolic blood pressure must be between 30 and 160"	422	pass
8	Visit one at 30 weeks	422, message contains "at or before 28 weeks"	422	pass
9	Visit two at the same week as visit one	422, message contains "later than visit one"	422	pass
10	Visit two at 28 weeks, upper boundary	201, assessment returned	201	pass
11	Visit two pulse and haemoglobin missing	201, assessment returned	201	pass
12	Visit two entered before visit one	409, message contains "Record visit one before visit two"	409	pass

4. Code quality practices

The package separates data access, parsing, cohort rules, partitioning, features, models, evaluation, the artefact and reporting into their own modules, and every constant lives in one frozen `StudySpec`. `FeatureSet` and `MetricStrategy` use abstraction and polymorphism. Every public class and function has a docstring and type hints; no function exceeds 50 lines and no file 300. Ruff and black run locally and in continuous integration, every output carries a checksum in `outputs/manifest.json`, and the README results are generated and tested. The history holds 90 commits with imperative messages.

5. Results and limitations

Result	Value (95 percent interval)
Primary, development: first visit added (C over B), average precision, logistic	-0.0040 (-0.0130 to +0.0025)
Primary, development: same, trees	-0.0066 (-0.0173 to +0.0024)
Smallest detectable difference, C over B, logistic	0.0110
Secondary, development: second visit added (B over A), ROC AUC, logistic	+0.0205 (-0.0066 to +0.0476)
Secondary, development: same, trees	+0.0345 (+0.0070 to +0.0627)
Sensitivity analyses with a C over B interval excluding zero	0 of 7
C over B, pooled multiple imputation	-0.0055 (-0.0151 to +0.0008)
Calibration slope, logistic, set C, development	0.98 (0.63 to 1.37)
Share of events in the top 10, 20 and 30 percent, development	21.8 percent (15.9 to 27.9); 34.5 percent (27.8 to 42.0); 47.9 percent (40.4 to 55.4)
Stage 2: largest average precision gain (gradient_boosting_weighted on C)	+0.0075 (-0.0036 to +0.0243); detectable 0.0193
Held out: primary, C over B, average precision	+0.0012 (-0.0221 to +0.0185), p 0.707
Held out: secondary, B over A, ROC AUC	+0.0405 (+0.0008 to +0.0805), p 0.046
Held out: served model average precision; ROC AUC	0.1023 (0.0532 to 0.1879); 0.5725 (0.4752 to 0.6721)
Held out: calibration slope of the served model	0.4475 (-0.4110 to 1.4021)
Held out: label sensitivity; specificity; correct when "Likely"	0.3415 (0.2059 to 0.4865); 0.7949 (0.7643 to 0.8226); 0.0843 (0.0471 to 0.1312)

Reading. Adding the first visit to the second did not improve prediction of recorded low birth weight by any amount this design could detect, on development records in either model family, in any sensitivity analysis or under multiple imputation, and the single held out evaluation agrees. The second visit over registration alone shows a modest gain in discrimination. None of the 28 prespecified Stage 2 experiments improved the model, so artefact 1.0.0 stays. On the held out records the label behaved as on development records, but overall discrimination was weaker and its ROC AUC interval includes 0.5, and the calibration slope is uncertain. With 41 held out events these intervals are wide.

Limitations. The predictors carry little information: fundal height is recorded only as normal or abnormal, the history counts are internally inconsistent, for example prior deliveries at or above the number of pregnancies (`docs/implementation_audit.md`), and birth weight is recorded to 0.1 kg. It is one public Tanzanian dataset (version 1 of the Zenodo record, frozen by checksum) with no facility or geographic identifier, so neither external nor geographic validation is possible, and the release contains no ethics approval or consent statement; a request was sent to the producers on 21 October 2026. The development partition holds 165 events, so only differences of roughly the detectable size could be seen. The outcome is birth weight under the care actually given, and nothing here shows that acting on a prediction would change it. The prototype runs on synthetic cases only and has not been validated for any pregnancy; a larger independent cohort is needed before any claim about use in practice.